

Chapter 4: The Network Layer–Data Plane

EXERCISE 1

Suppose two packets arrive to two different input ports of a router at exactly the same time. Also suppose there are no other packets anywhere in the router.

- a) Suppose the two packets are to be forwarded to two different output ports. Is it possible to forward the two packets through the switch fabric at the same time when the fabric uses a **shared bus**?

No. It is not possible to forward two packets to different

output ports at the same time using a shared bus fabric, because a shared bus can only transmit packet once at a time, meaning only one packet can be forward at any moment.

- b) Suppose the two packets are to be forwarded to two different output ports. Is it possible to forward the two packets through the switch fabric at the same time when the fabric uses **switching via memory**?

NO. In the text book, only one memory read/write can be

done at a time over the shared system bus

- c) Suppose the two packets are to be forwarded to the same output port. Is it possible to forward the two packets through the switch fabric at the same time when the fabric uses a **crossbar**?

No, It is not possible to forward both packets at the same time through the router to one output port, because although the crossbar allows multiple packet through, they cannot be at the same port and the time.

EXERCISE 2

Consider a datagram network using 32-bit host addresses. Suppose a router has four links, numbered 0 through 3, and packets are to be forwarded to the link interfaces as follows:

Destination Address Range	Link Interface
11100000 00000000 00000000 00000000 through 11100000 00111111 11111111 11111111	0
11100000 01000000 00000000 00000000 through 11100000 01000000 11111111 11111111	1
11100000 01000001 00000000 00000000 through 11100001 01111111 11111111 11111111	2
otherwise	3

- a) Provide a forwarding table that has five entries, uses longest prefix matching, and forwards packets to the correct link interfaces.

Prefix (CIDR)	Link Interface
11100000 00	0
11100000 01000000	1
11100000 01000001	2
Default Route (0.0.0.0/0)	3

b) Describe how your forwarding table determines the appropriate link interface for datagrams with destination addresses:

11001000 10010001 01010001 01010101

11100001 01000000 11000011 00111100

11100001 10000000 00010001 01110111

① 11001000 10010001 01010001 01010101, this address doesn't match any of the CIDR blocks above, so it should be link Interface 3

② 11100001 start with 11100000 01000001, which is Interface 2

③ 11100001 10000000 does not match any CIDR in the table so it's Interface 3

EXERCISE 3

Consider a router that interconnects three subnets: Subnet 1, Subnet 2, and Subnet 3. Suppose all of the interfaces in each of these three subnets are required to have the prefix 223.1.17/24. Also suppose that Subnet 1 is required to support at least 60 interfaces, Subnet 2 is to support at least 90 interfaces, and Subnet 3 is to support at least 12 interfaces. Provide three network addresses (of the form a.b.c.d/x) that satisfy these constraints.

223.1.17/24 (Network ID)	network address	subnet mask
Subnet 1	223.1.17.0/26	/26
Subnet 2	223.1.17.128/25	/25
Subnet 3	223.1.17.192/28	/28